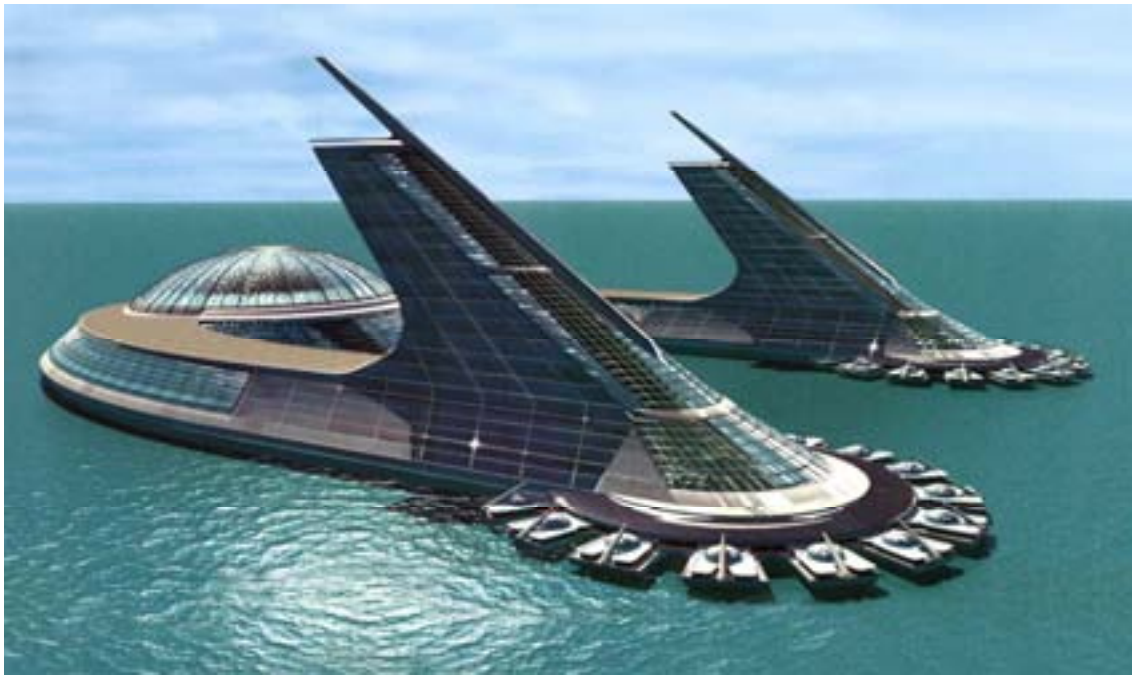
# DESIGNING THE FUTURE



BY

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# INTRODUCTION

## A Future by Design

Are you prepared to design the future?

Although many of us feel we can prepare for our future by thinking, acting, and learning using present methods and values, nothing is farther from the truth – especially in today's rapidly changing world. A newborn child enters a world not of his or her own making. Each succeeding generation inherits the values, accomplishments, hopes, successes, and failings of previous generations. And they inherit the results of the decisions made by those generations.

For the hundreds of thousands of years of human existence when technologies were simple or non-existent, this may have had little impact on human life and the earth that sustains it. Each generation of hunters and gatherers, then plowmen and pioneers, passed on tools to the next generation to help them survive. Change from one generation to the next was slow and hardly noticeable. In those days there was little understanding of science and how things worked, and explanations were not scientific.

This is no longer the case in today's high-tech world where a change that affects millions may happen in a matter of seconds. A child born today inherits a world vastly different from that of its parent's generation, let alone that from centuries ago. Previous generations left a legacy of, exploitation, occupation, and irrelevant values that present great challenges, but also opportunities to the people of today.

The application of scientific principles, for better or worse, accounts for every single advance that has improved people's lives. Important documents and proclamations have been issued granting rights and privileges to members of societies, but at the heart of human progress – or destruction – is the rock-solid foundation of science.

For generations past it was impossible to direct the future much beyond the present moment, and forecasts of the future were based on non-scientific methods. Prophets and sages presented visions of the future based on dreams, hallucinations, religious fervor, divination of animal

parts, crystal balls, etc. Some may even have been accurate, but this was more because of luck than because of any direct channel to the supernatural.

Now satellites circle the globe beaming down information in fractions of a second about everything that impacts our lives. This information is very valuable for projecting weather patterns, high and low points, geological hot and cold spots, where people live, and the warming of the planet. This has given us, for the first time, the ability to monitor the health of the planet, which many scientists see as in serious, if not critical, condition.

In a single day, trillions of bits of scientific data zip through cyberspace at light-speed, making a high-tech civilization possible. While physical science and technology silently direct much of the action, millions of people around the globe still practice pseudo-science, using fortune-tellers, seers, and philosophers for their daily direction. Many world leaders regularly consult psychics, mediums, and astrologers for guidance in decisions that determine the fate of millions.

Present human activity and its consequences does not *have* to be shaped by the needs and values of our ancestors. In fact, it *must* not be. For instance, armed conflict between nations is still seen by many as the only way to settle differences. It is especially promoted by those who profit handsomely from the sale of armaments. This is now totally unacceptable and dangerous because of war's extreme human and environmental costs.

A militant viewpoint is obsolete once we view the world as a whole interrelated system with all its people as one family. Managing accelerating changes in technology and managing ourselves require new outlooks and approaches. This is now both necessary and possible because of technological change.

These lessons are designed to challenge the reader to direct the future; not just one's own, but that of society in general; and not just for one's own generation, but for those to follow. Not only is science making it possible, it is now vital.

## YOUR CHALLENGE

The future does not just happen. Except for natural events like earthquakes, it comes about through the efforts of people and is determined largely by how well informed people are. You can play a role in the shaping of tomorrow's world by asking yourself questions like, "What kind of world do I want to live in?" and "What does democracy mean to me?" There are many other options of organization for the future than those typically discussed today.

Here is a scenario for you to consider: Suppose you were called upon to redesign planetary civilization without any limitations based on how things are done today. The goal is to help rid the world of war, poverty, hunger, and environmental degradation, and to create the best world for all inhabitants, given the resources at hand, for the longest period of time.

Remember, you are free to rearrange society in any way you think will work. The only limitation is that your social design must factor in the carrying capacity of the planet, which means the resources have to be sufficient to support life on the planet.

You can rearrange the entire civilization to make what you consider the best of all possible worlds, bearing in mind that any unmet need for any segment of the population reduces the standard of living for all. This may include not only environmental protection, but also city design, transportation, interpersonal relationships and the restructuring of education, if you feel it is necessary.

The options are limitless. Would you have separate nations? Would you have an international advisory board? How would you manage and distribute the resources of the world to accommodate the needs of all? Would you use the scientific method to make decisions, or rely on politics or mysticism? How would you handle differences in religious beliefs? You may even consider another system of distribution that doesn't use money as a medium of exchange.

On a personal basis, would you seek a position of advantage over others? Would you claim a bigger house, a more luxurious car, or high-definition TV? On what basis would you say you deserve these things? Or that others

don't deserve them? Your skill level? Your investment of time and/or money?

Remember, if you force any predetermined set of values on other nations, or others in your own nation or neighborhood for that matter, you will generate bad feelings. How would you prevent political corruption? Would you declare universal laws and treaties? Would you use military and police methods for enforcement? Would you declare all resources the common heritage of all nations?

In order to accomplish this task one must be free of bias and nationalism, and reflect those qualities in the design of policies. How would you approach that? This is a difficult project requiring input from many disciplines.

These are some of the problems we must consider when thinking about such a task. It can be a fresh approach, unburdened by past or traditional considerations, religious or otherwise, but always keeping in mind for whom this society is to be designed. Feel free to transcend present realities and reach out for new and creative ideas.

## Chapter One

### From Yesterday to Tomorrow

**A little background before considering this challenge:**

The lives of most men and women are blighted by problems they cannot solve. Many events in our lives are the result of things beyond our control. While it is comforting to think, "I'm in charge," in truth most changes effected by individuals are very limited in scope. People usually blame themselves or "fate." However, when two cars collide at an intersection, should we blame the individual drivers, "fate," or the way transportation is engineered so that it permits collisions in the first place? Are we, as an individual, in charge if the automobile colliding with us is the product of poor design?

In 2005 there were 43,200 thousand deaths in the US from car accidents, plus hundreds of thousands of injuries. But consider another way we get people from one place to another – the elevator. How many people have been killed in collisions between elevators? These devices carry millions of people every day without a single mishap because of their intelligent design. How might highway transportation be similarly arranged?

If you believe that transportation should be designed so that it is almost impossible for anyone to be killed or injured in a collision, these lessons are for you. If you believe that scientific investigation can find out how to restructure society to give each individual a greater opportunity for self-realization and fulfillment, then you will probably appreciate these ideas.

To get the most out of these ideas you will have to blend open-mindedness with skepticism. It is hard enough to face the problems of our own time; it is even more difficult to understand the fantastic and shocking changes that may occur in the future.

Suppose an intelligent man in New York a hundred years ago sat down one evening with a book predicting life a century later. He would refuse to believe that almost everyone in 2006 would be able to operate a horseless carriage that could zip about at 60 miles per hour or more. He may have thought the designers had gone too far.

He would smile smugly at the ridiculous prediction of man-made flying machines traveling faster than the speed of sound. The thought of sending pictures and sounds throughout the world instantly would have sounded impossible to such a person a hundred years ago. It would have seemed incredible to him that war would develop to the point where one small bomb, directed in real time from half way around the world, could destroy an entire city with pin-point accuracy. Our early 20th century gentleman would have been alarmed that part of his wages could be withheld to provide for retirement.

At this point let us leave our gentleman muttering to himself about the world moving too fast and a future that has gone too far.

Are we any more flexible or farsighted today? In order to design a future of positive change, we must first become expert at changing our minds. The differences between the nineteenth and the twentieth centuries will probably be small compared to the changes that will take place in the rest of our own century.

Students will understand these ideas best if they can see today as a stepping-stone between yesterday and tomorrow. They will also need sensitivity to the injustices, lost opportunities for happiness, and deadly conflicts that characterize our twenty-first century civilization.

We have no crystal ball for the rest of the twenty-first century. We want you to feed these ideas into your own mental computer and experience. You may find even better ideas that can play a part in molding the future of our civilization. In the rest of these pages, we'll explore unfamiliar, alarming, exciting, and attainable possibilities for the design of the future.

**A Crisis That Needs to be Addressed**

One would think that with our technology we could eliminate most social ills. Couldn't modern technology supply enough food, clothing, shelter, and material goods for all if used intelligently? What is stopping us from achieving this? Technology is racing forward but our societies are still based on concepts and methods devised centuries ago. We still have a society based on scarcity and the use of money. We still have thinking patterns based on old structures used in western Asia several thousand years ago. We are trying to adjust to the rapid advances in technology with obsolete values that no longer work in today's world.

Because of tremendous advantages given corporations by lawmakers who owe them their positions, monopolies are gaining more control. The comforting assumption that "I can make a difference" is farther and farther from reality. Fewer corporations own more and more companies. Many of the same people sit on the boards of various major corporations besides their own. The corporations that own car and aircraft companies may also own food, radio, TV stations, magazines, pharmaceuticals, manufacturing, and armament companies. Ten major lending institutions control virtually all the credit cards in the U.S. The wealth and influence of these corporate elite can not be equaled or countered by the workers who enabled them to acquire such wealth. With media companies today owned and sponsored by large corporations, it is difficult to know whether the news can be trusted.

According to many polls, a majority of scientists think that the human race is on a "collision course" with nature, that all of Earth's ecosystems are suffering, and that the ability of the planet to sustain life is in serious jeopardy. (1) There is a threat of rapid global climate change that will certainly have profound consequences. The pollution of rivers, land, and the air we breathe threatens our health. We are destroying non-renewable resources like topsoil and the ozone layer instead of using these resources intelligently.

We face common threats that transcend national boundaries: overpopulation, energy shortages, water scarcity, economic catastrophe, the spread of uncontrollable diseases, and the technological displacement of people by machines, to name a few. Eight hundred and fifty two million people across the world are hungry. Every day, more than 16,000 children die from hunger-related causes- one child every five seconds. (1) World wide more than 1 billion people currently live below the international poverty line, earning less than $1 per day. (2) A very small percent of the people own most of the world's wealth and resources. The gap between the rich and poor is widening. In the US as of 2002, the average CEO made 282 times as much as the average worker. (3) In 2005 the compensation of CEOs of major U.S. corporations rose 12% to an average of $9.8 million per year. Oil company CEOs did even better with raises that averaged a whopping 109% to $16.6 million per year. Meanwhile, workers' salaries barely kept up with inflation in most industries and occupations across the U.S. In Oregon, minimum wage workers saw their pay rise by a modest 2.8% to $15,080 per year.

What has been handed down to us does not seem to be working for the majority of people. With the advances in science and technology over the last two hundred years, you may be asking: "does it have to be this

way?" With the observable fact that scientific knowledge makes our lives better when applied with concern for human welfare and environmental protection, there is no question that science and technology can produce abundance so that no one has to go without. But the misuse and abuse of technology seems to make things worse.

The problems we face in the world today are mostly of our own making. We must accept that our future depends on *us*. While the values represented by religious leaders over the centuries have inspired many to act in a socially responsible manner, others have gone to war over their differences in religious beliefs. Hopes for divine intervention by mythical characters are delusions that cannot solve the problems of our modern world. The future of the world is our responsibility and it depends upon decisions we make today. We are our own salvation or damnation.

The shape and solutions of the future rely totally on the collective effort of people working together. We are all an integral part of the web of life. What affects other people and the environment has consequences in our own lives as well.

What is needed is a change in our sense of direction and purpose -- an alternative vision for a sustainable new world civilization unlike any in the past. Although this vision is highly compressed here, it is based on years of study and experimental research.

These writings offer possible alternatives for striving toward a better world. It arrives at decisions using the scientific method. Like any new approach, it requires some imagination and a willingness to consider the unconventional in order to be appreciated. Remember that almost every new concept was ridiculed, rejected, and laughed at when first presented, especially by the experts of the time.

That's what happened to the first scientists who said the earth was round, the first who said it went around the sun, and the first who thought people could learn to fly. You could write a whole book, and many have, just on things that people thought were impossible up until the time they happened. Imagine going to the moon, for example! Your great-grandparents would have laughed at such a notion! Such notions were the ramblings of science fiction writers. Many forward thinking people have been locked up and even executed for saying such things as the earth wasn't the center of the universe.

Those who fought for social justice and change had even greater difficulties. People advocating change were beaten, abused, put in

prison, and brutally murdered. For example, Wangari Maathai, who was awarded the 2004 Nobel Peace Price on December 10, 2004 was tear gassed, beaten unconscious, and imprisoned for fighting against deforestation in Kenya, Africa. Dianne Fosse, the naturalist who actively strived to protect declining gorilla populations from poachers, was found hacked to death in her hut. Unfortunately she did not provide for the needs of the poachers. Any number of volumes could be written on the hardships endured by those who sought change that threatened the status quo.

(1)**The world** hunger **problem: Facts, figures and** statistics
**http://library.thinkquest.org/C002291/high/present/stats.htm**

(2) Hunger Report 2004. Bread for the World Institute
Hunger **Facts: International**

(3) Capital Connection
**http://www.oraflcio.org/cgi-bin/display.cgi?page=CapConnect42505**

## Chapter Two

## All Things Change

In our dynamic universe all things change, from the farthest reaches of outer space to the movement of continents. Change occurs in all living and nonliving systems. The history of civilization is the story of change from the simple to the more complex. Human ingenuity and invention bear witness to this fact. No system can remain static for long; most of the monarchies have been replaced by other forms of government and societies based on the will of the people, not kings, have evolved. Unfortunately, the changes are not always for the best.

Although we accept the inevitability of change, humans meet it with a lot of resistance. In most cases, change threatens those in positions of advantage and for the most part they are there in the first place to keep things the way they are. This is true for any society, whether the power structure is religious, military, socialist, capitalist, communist, fascist, or tribal. The leaders will attempt to hold back change. Sometimes, even when conditions are terrible for the majority of people, the people themselves may resist change because there is comfort in the familiar. We refer to them as the un-appointed guardians of the system.

But no matter how much people resist, human civilization is no exception to the fact of change. Change occurs in all social systems, and is the only constant. We can be sure that the history of humankind is one of change.

Yet at every turn, vested interests (those who have the most to gain in keeping things the way they are) oppose even technological changes. For example, earlier in the twentieth century, defenders of the horse-mounted cavalry delayed development of the tank. So entrenched was this tradition that when Germany invaded Poland in 1939, their tank division faced Polish soldiers still mounted on horseback.

It was obvious that the horse soldiers did not stand a chance. The development of aircraft threatened tank divisions. Then pilots and aircraft designers fought to hold back the development of guided missiles. The missile men fought to hold back the development of laser weapons. And so it goes.

If we wonder why we are still faced with many of the same problems our ancestors had when our technological capabilities so surpass theirs, we must consider that we have been here such a short time that we could almost be called “newborn.” If you were to use a twenty-four hour clock to represent the time since life began on earth, it would show that humans have only been in existence since the last minute of the twenty-fourth hour. Only during the last few seconds of the last minute have modern humans begun to use scientific methods to find out the most effective ways of getting things done. We are just now beginning to hit our stride. More new knowledge has been created from the beginning of the twentieth century to the present than in the previous billion years. Change is almost everywhere.

If life at times seems bewildering - if you feel pulled in many directions, if you find that no matter what you do, you still have sticky problems, if you find that our economic, political, and social ways of doing things sometimes create more difficulties than they solve - then you are simply playing your part in suffering through the present transitional phase of our civilization.

## Chapter Three

## Using the Scientific Method

### What Have We Here

Until scientific inquiry came of age, human beings could not comprehend their relationship to the physical world, so they invented their own explanations. These explanations tended to be simplistic and in many cases harmful. For example, if one knows a tidal wave is approaching and chooses to stay and pray for deliverance rather than leaving, this could be detrimental to his/her survival. People used to believe that plagues and diseases were retributions of an angry God, but the scientific method found that many diseases were carried by rats and lice, and caused by germs.

It is not that scientists are close-minded regarding these issues—it's just that their acceptance of ideas requires more sophisticated standards and methods of inquiry.

The scientific method helps to diminish biases, prejudices, and preconceived notions. The method requires that statements be verified and that researchers find out through experimentation just what works and what doesn't. Scientists ask the question "what do we have here?" and then they proceed to do experiments to determine the nature of the physical world.

This process requires that experiments be verified by others who must get the same results. One of the major developments in science was the realization that we can not acquire answers to problems just intuitively. It requires painstaking laborious effort and time to find solutions and answers. Often many failures come before any new findings.

### The Language of Science

The communication of ideas and information usually starts with language, but when you see how misunderstood you may be in daily life, you know this can be a confusing task. Our everyday language evolved through centuries of cultural change and unfortunately, it is difficult to resolve conflicting ideas by using it. In most instances, because of varying

backgrounds and life experiences, the same word can have different meanings for different people. One's thoughts may be interpreted differently by others even when using the same language.

But there is a language that is easily understood by many, even in different parts of the world. This language has a *high degree of physical correlation with the real world*. There is little to no confusion in it. In different scientific fields such as engineering, mathematics, chemistry, and other technical areas, we have the nearest thing to a descriptive universal language that leaves little room for one's own unique interpretation.

For instance, if a blueprint for an automobile is given to any technologically developed society anywhere in the world, regardless of political or religious belief, the finished product will be the same. This language was deliberately designed as a more appropriate way to state a problem. It is nearly free of vague interpretations and ambiguities.

Many of the great technical strides made in our modern day would have been unattainable without this improved communication. Without a common descriptive language, we would have been unable to prevent diseases, increase crop yields, talk over thousands of miles, or build bridges, dams, transportation systems, and the many other technological marvels of this computerized age.

Applying and understanding General Semantics is essential to improving communication. Semantics has been defined in many different ways. Briefly, it is an attempt to improve communication through the careful use of language. For example, terms like "Arab", "Jew", or "Irishman" have slightly different meanings to different people. Similar words have various meanings depending on differences in background and experience. This also applies to words like understanding, conscience, democracy, reality, love, etc. To have intelligent discussions using certain words, it is essential to ask what one means by the words being used. If one wishes to communicate in a meaningful way, it is best to have people define their terms. Semantics is just one instrument that may help improve communication. A useful book for this subject is *Tyranny of Words* by Stuart Chase.

**Can we apply the Methods of Science to how we Design our Society?**

The discovery of scientific principles enables us to validate and test many proposals. If someone claims that a certain structural element can support a specific number of pounds per square inch, this statement can be tested and either accepted or rejected based on the test results. It is this

testing which enables us to design and construct bridges, buildings, ships, aircraft, and other mechanical wonders.

Almost everyone you know will go for the scientific way when it comes to surgery, airplane rides, or building things like skyscrapers, bridges, and cars. Over the centuries, we seem to have developed a consensus that when it comes to matters of personal safety, we will go with the science rather than the magic. Why is that? Probably because it works, and everybody can see that it does.

Then why don't we do that when it comes to planning our societies: our cities, transportation systems, agriculture, health care, and so on? If you thought we were *already* doing all those things scientifically, look again! If science has a lot to do with what works, then clearly there's much about today's social and economic setup that isn't scientific, because things aren't working very well for a majority of the world's population or the environment. If they were, war, poverty, hunger, homelessness, pollution, etc., would not be so prevalent today. Unfortunately our social structures evolved with no overall global planning.

One condition for the assignment of redesigning society is that your social design must live within the carrying capacity of our planet. This means our resources have to support life on the planet for everyone. This will certainly require scientific methods of evaluation.

If one wishes to put a person on the moon, one cannot just build a rocket and head for the moon. We must first test what forces the human body can stand. We would put a person in a centrifuge to see just how many "G's" the body can withstand. We would put a person through a barrage of tests. For instance, we would test to see how the body functions in a gravity- free environment and the effects on human health. We would also need to have information about survival possibilities on the moon; for example, is there water, air, an acceptable temperature range, etc.

In much the same way we must look at the entire planet as a whole and ask "what have we here?" We want to apply this same intelligent method of planning using a scientific system of Earth sciences for planetary survival. The degree to which we *don't* apply this scientific method to the way we live on Earth may very well determine the unnecessary amount of suffering that will occur.

How do we do this?

## Chapter Four

## Existing Myths

### The Rule of Law

Many people feel that we need the rule of law to eliminate our problems. But is it more laws that we need? We have many laws – thousands upon thousands of them - but they are constantly being broken.

For instance, there are thousands of laws against stealing. But if we examine this more closely and look at the statistics, we find that a few people control most of the earth's resources. Most people have insufficient money to purchase even the barest of necessities. How can we think that without altering these conditions, the passing of a law will prevent stealing? This is even more difficult when advertising makes products so enticing. Almost unknowingly, people in the US are exposed to over 2500 advertisements a day.

Even a peace treaty cannot prevent another war if the underlying causes are not dealt with. Laws for international cooperation do not deal with the reasons we need the laws -- they just tend to freeze things as they are. Regardless of treaties, nations that have conquered land all over the world by force and violence still retain their positions of territorial and resource advantage. Treaties are only a band-aid on the problems and usually only work to postpone conflict for a short time.

Perhaps what is needed are different people in government, ethical people who are concerned about others. Maybe they will wipe out corruption and work towards everyone's well-being. But even if the most ethical people were elected to high position and we ran out of resources, there would still be lying, cheating, stealing, and corruption. *It is not ethical people that are needed but rather a way of intelligently managing the Earth's resources for everyone's well-being*.

### Examining the Conditions that Cause the Problems

Perhaps the problem lies elsewhere than in the enactment of more laws or appointing ethical people to government. Perhaps we should look at how we currently obtain and distribute the goods that we need.

This is done by "earning" money, either by exchanging one's time, skills, and efforts, or by "investing" in the financial system with the thought of getting more money in return, and exchanging that money for goods and services. This may have been a good method in the past when goods were scarce and technology was in its infancy, but today our advanced technologies could be the tools for a very different scenario.

If we look at things scientifically, there is more than enough food and material goods on Earth to take care of all people's needs – if managed correctly. There is enough to enable everyone to have a very high standard of living with the intelligent use of technology, resources, and technical personnel. When we say the use of technology, we mean technology that is not harmful to people or the environment and doesn't waste time and energy.

Consider this: when there is a recession and people have little money to buy things, isn't the Earth still the same place? Aren't there still goods on the store shelves and land to grow crops? It is just the rules of the game that we play by that are obsolete and cause so much suffering.

The existence of money is hardly ever questioned or examined, but let's consider our use of money. Money itself does not have any value. It is just a picture on a cheap piece of paper with an agreement among people as to what it can buy. If it rained hundred dollar bills tomorrow, everyone would be happy except the bankers.

There are many disadvantages to using this old method of exchange for goods and services. We will consider just a few here and let you add to this list on your own.

1. Money is just an interference between what one needs and what one is able to get. It is not money that people need, it is access to resources.

2. The use of money results in social stratification and elitism based primarily on economic disparity.

3. People are not equal without equal purchasing power.

4. Most people are slaves to jobs they do not like because they need the money.

5. There is tremendous corruption, greed, crime, embezzlement, and more caused by the need for money.

6. Most laws are enacted for the benefit of corporations, which have enough money to lobby, bribe, or persuade government officials to make laws that serve their interests.

7. Those who control purchasing power have greater influence.

8. Money is used to control the behavior of those with limited purchasing power.

9. Goods such as foods are sometimes destroyed to keep prices up; when things are scarce prices increase.

10. There is tremendous waste of material and strain on available resources from superficial design changes for newer later fads each year in order to create continuous markets for manufacturers.

11. There is tremendous environmental degradation due to the high cost of better methods of waste disposal.

12. The Earth is being plundered for profit.

13. The benefits of technology are only distributed to those with sufficient purchasing power.

14. Most important, when the corporation's bottom line is profit, decisions in all areas are made *not* for the benefit of people and the environment, but primarily for the acquisition of wealth, property, and power.

**Next Phase in Social Development.**

What is it that we all have in common? Where should our priorities lie? All nations and people, regardless of political philosophy, religious beliefs, or social customs, depend upon natural resources; we all need clean air and water, arable land for food, and the necessary technology and personnel to maintain a high standard of living. Perhaps we should update the way society works so everyone on earth can take advantage of our technological ability to maintain a clean environment and a high standard of living. There is not enough money to begin to pay for this type of change, but there are more than enough resources on Earth to create it.

To review: the Earth has abundant resources and our practice of rationing these resources through the use of money is an outdated method which causes much suffering.

It is not money that we need but the *intelligent management of the earth's resources for the benefit of everyone*. We could best work towards achieving this by using a **resource based economy**.

**Recourse-based Economy**

This is a very different concept than anything else put forth today. To put it simply, a resource-based economy uses resources rather than money, and people have access to whatever they need without the use of money, credits, barter, or any other form of debt or servitude. *All of the world's resources are held as the common heritage of all of Earth's people.*

The real wealth of any nation is not its money, but the developed and potential resources and the people who work toward the elimination of scarcity for a more humane society.

If this is still confusing to you consider this: If a group of people were stranded on an island with money, gold, and diamonds, but the island had no arable land, fish or clean water, their wealth would be irrelevant to their survival.

What if all the money in the world suddenly disappeared? As long as topsoil, factories, and other resources still remained, we could build anything we chose to build and fulfill our material needs. The bottom line is that money is not what people really need; rather, it is access to the necessities of life.

In a *resource-based economy,* resources are used directly to enhance the lives of our total population. In an economy based on resources rather than money, we can easily produce all of the necessities of life and provide a very high standard of living for everyone.

## Chapter Five

## From one System to Another

### The Transition - Signs of the Times

Most people don't begin to look for an alternative social arrangement until theirs no longer works for them. A change from a system as entrenched in our culture as money will most likely require a collapse of the current system. Some things that are happening today could be signs that this collapse is already underway:

The industrialized nations of the world are installing more and more automated technology in order to compete with low prices in the global economy. The result of this new technology is that more and more people loose their jobs and can't take care of themselves and their families. With automation and cybernation used to their fullest potential, machines replace not only industrial workers, but also most professionals. Consequently, fewer people are able to buy the products that automated factories turn out.

A continuous outsourcing of jobs and manufacturing plants to overseas for cheaper labor, fewer environmental restrictions, and other benefits may seem good in the short run but will ultimately prove disastrous. It is likely that the loss of income for a majority of unemployed will become so great that they will lose their homes and possessions.

A number of scientists claim that by the year 2030, there will be a drastic shortage of easily extracted oil. Oil may not run out, but it may become monetarily and then physically impractical to extract it. Eventually, it will require more energy to drill for it and refine it than is practical. It is likely the same will happen with natural gas, only more rapidly.

These developments will create tremendous social and environmental disruptions as businesses scramble to protect their profit margins and exploit more of the earth's land, water, and natural resources. It may take the failure of the debt/money system for the majority of people to loose confidence in it. Then they can examine seriously how a global resource based-economy would operate, and envision what life would be like in

such a society. In the chapters that follow, we will glimpse the processes involved in adapting ourselves to this new way of life.

# Chapter Six

# Future By Design

## Emerging Into a Saner Future

### First Steps

To begin implementing a resource-based economy, social designers must utilize the scientific method and pose the question "What do we have here?" With the requirement that all be provided for in the most efficient, comfortable, and enduring manner possible, the first priority is making a purely technical assessment of basic needs of the total global population. The amount of housing, food, water, health care, transportation, education, and such needed, must be compared to the available resources the planet has to offer. This has to be balanced with the needs of other species that make up the web of life on Earth.

*The main objective is to overcome scarcity and provide for the needs of all the world's people.* In order to create a workable and sustainable civilization as quickly as possible, we need vast amounts of energy. What is desperately needed is an energy development strategy on a global scale, requiring a joint venture of international planning on a level never before achieved.

### Energy

One of the most useful measures of the development of civilization is the amount of available energy per person. To a large extent the degree of physical comfort you enjoy today correlates with the energy at your disposal. Imagine the paralyses that would occur if your electricity and gasoline supply were cut off, and you had to use your own muscles to get things done.

The resource-based economy quickly goes to work on clean sources of energy. This is only possible when there are no more monetary limitations in the way of accomplishing or providing what's needed. With the restrictions of profit, property, and scarcity eliminated, research labs would quickly begin working together and sharing information freely. There would be no need for patents or proprietary information since the end goal is not to make money in order to continue working, but to achieve

results that are freely and quickly available to the planet's entire population.

This is a project that many people would be eager and grateful to work on, when the results immediately benefit all people. Interdisciplinary teams of qualified personnel, in line with the project's requirements, will work on energy and automated systems to produce and supply goods and services on a massive scale. Even the university students will help participate in arriving at fast methods of solving these problem.

These can be the armies of the future, a large peaceful mobilization to restore and preserve the earth and its people. This has never been done before and can only be done when money is no obstacle. The question is not do we have the money, but do we have the resources and means to accomplish this new direction.

During the transition from one system to another, scarcity regions are provided with heat concentrators for cooking and sterilizing water. Foods for those areas are dehydrated and compressed to save shipping space. The packaging is biodegradable and may double as non-contaminating fertilizers. Regions without arable land will use hydroponic farms, land-based fish farms, and sea farming. To conserve energy during the transition, instead of each family preparing food, there are food distribution centers with food shipped directly to homes and restaurants. These massive methods of supplying goods and services are applied throughout the world.

Vast sources of energy will be explored and developed. These include wind, wave and tidal action, ocean currents, temperature differentials, falling water, geothermal, electrostatic, hydrogen, natural gas, algae, biomass, bacteria, phase transformation, and thermionics (the conversion of heat into electricity by boiling electrons off a hot metal surface and condensing them on a cooler surface). Additionally, there is the potential of Fresnel lenses to concentrate heat.

Fusion energy is the same energy that drives the cosmos and the stars. When we learn how to harness it, the world's energy problems will be solved forever, without any detrimental effects or dangerous toxic materials to be disposed of. The only residue would be the clean ash of helium.

Oceanographers told us in the late twentieth century that if we tapped the vast energy potential of the world's oceans occupying 70.8% of the

earth's surface, we could easily meet present and future energy needs for millions of years to come.

A key element in the design of cities in the resource-based economy is the embedding of all necessary energy harnessing within the structure of the city itself. This will be further explained in the "City" section.

Another vast untapped energy option is the development of piezoelectric materials, or laminated systems inside cylinders, activated by the rise and fall of the tides.

Geothermal power, or power extracted from the heat of the earth, is being used throughout the world with tremendous success. Scientists predict that if we develop and harness only 1% of the geothermal energy available in the crust of the earth, our energy problems would be eliminated. With no monetary restriction in the resource-based economy, society would have the chance to prove those scientists right.

Geothermal energy can supply more than 500 times the energy contained in all the world's fossil fuel resources while reducing the threat of global warming. Geothermal power plants produce very little pollution compared to fossil fuels, and emit no nitrogen oxide or carbon dioxide. A relatively small area of land is required for the power plant itself. Without oil and natural gas companies controlling a monetary economy, geothermal power would become the most economical and efficient way to heat and cool buildings. If we were to apply just one tenth of what is currently spent on military equipment to the development of geothermal generators, we could have long ago solved our energy shortages.

In areas such as Iceland, geothermal energy is used to grow plants year round in enclosed areas. In the resource-based economy using this method, enormous amounts of fresh vegetables can be cultivated in all seasons. A similar process can be used for fish farming and in regions where heating and cooling are needed. Massive underwater structures could run a portion of the flow of the Gulf Stream through large turbines to generate clean electric power. The turbines would have a centrifugal separator and deflectors to prevent harm to marine life.

A land bridge or tunnel across the Bering Strait between Asia and North America could generate electrical power and collect and process marine products. Beneath and above the ocean surface would be tunnels to transport passengers and materials. Pipelines could bring fresh water from melting icebergs to other parts of the world. This structure

would not only provide a physical link between continents, but would also serve as an avenue for social and cultural exchange.

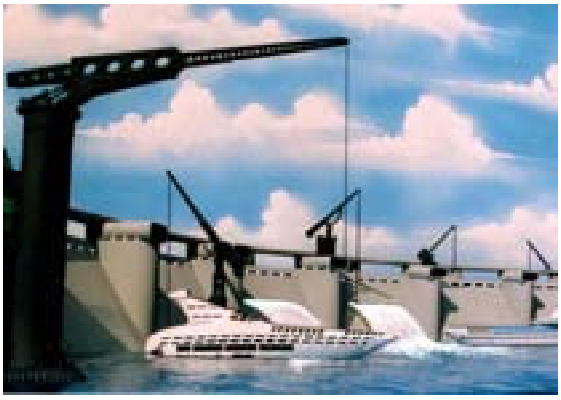
In our resource-based economy, there are comprehensive studies of the environmental and human impact before starting any large projects. The major concern is to protect and restore the environment for the benefit of all living creatures in the community of life. The purpose of the construction and development of these power projects is to free human beings from unnecessary laborious tasks. In order to achieve this society, we will need to automate most jobs as quickly as possible.



## Harnessing the Gulf Stream

These underwater structures divert a portion of the flow of the Gulf Stream and other currents through turbines to generate clean electric power. The turbines are slow turning and would have centrifugal separators and deflectors to prevent harm to marine life.

## Bering Strait Dam



A major development in the future could be the construction of a land bridge or tunnel across the Bering Strait. The primary function of this span would be to generate electrical power and house facilities for collecting and processing marine products. Beneath and above the ocean surface would be tunnels to transport passengers and materials. Pipelines to conduct fresh water from melting icebergs to other parts of the world may also be incorporated. Not only could this structure provide a physical link between Asia and North America, it could also serve as an avenue for social and cultural exchange.

## Geothermal Energy Plants



Geothermal energy, with refinements in conversion technologies, can take a more prominent role in reducing the threat of global warming. Readily available in many regions throughout the world, this source alone would provide enough clean energy for the next thousand years.